22 Chronic Cracks and Fractures

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CHAPTER OUTLINE

Impact Trauma
Diagnostic Challenge
Fracture Mechanics
Cracked and Fractured Cusps
Definition
Diagnosis
Patient History
Clinical Manifestation
Diagnosis
Etiology
Treatment Planning
Cracked Cusp
Fractured Cusp
Cracked and Split Teeth
Definition
Diagnosis
Patient History
Clinical Manifestation
Diagnosis
Etiology
Treatment Planning
Cracked Tooth
Split Tooth
Vertical Root Fracture
Definition
Diagnosis
Patient History
Clinical Manifestations
Diagnosis
Etiology
Natural Predisposing Factors
Iatrogenic Predisposing Factors
Treatment Planning
Summary

The presence of cracks or fractures in teeth can be difficult to diagnose and clinically manage. They may present with no symptoms or the symptoms may be vague or specific. Symptoms alone are often insufficient for making a definitive diagnosis and the radiographic interpretation can be inconclusive. Although the clinical management of the crack or fracture depends on its extent, often the extent of the crack or fracture cannot be clinically determined.

According to the 2016 version of the American Association of Endodontists (AAE) *Glossary of Endodontic Terms*, a *crack* is defined as "a thin surface disruption of enamel and dentin, and possibly cementum, of unknown depth or extension." A *fracture* is defined as "a split or break in bone, cartilage or tooth structure. When involving tooth structure, it is a disruption of the enamel and/or dentin and/or cementum that is assumed to be of greater depth than a crack, of unknown depth or extension, visible or not visible clinically or radiographically, and with separated or unseparated segments."[3a]

Impact Trauma

Traumatic injury to the tooth can be accompanied with a fracture in the crown or root system. These types of traumatic fractures mainly occur in the anterior segment of the mouth and are elaborated on in Chapter 21). In contrast, the cracks and fractures described in this chapter are often not associated with a traumatic event. These cracks and fractures are frequently the result of an accumulating, unobserved trauma resulting from either normal or excessive occlusal forces that are repetitively applied[22,35,46,48,71,72,91] without the patient's awareness. Specifically discussed in this chapter will be crown originating fractures and longitudinal fractures in the roots.

Diagnostic Challenge

Three categories of cracks and fractures are discussed in this chapter: crown originating fractures, split teeth, and vertical root fractures (VRFs). Each is often undiagnosed or misdiagnosed for a relatively long time.[16,19,81] The astute clinician can usually diagnose a pulpitis, apical periodontitis, or an abscess after appropriate diagnostic tests and clinical assessments are performed. However, because of the wide variety of clinical presentations from cracks and fractures, the diagnosis is less straightforward.[16,18,79] To complicate the diagnosis, most cracks and fractures exhibit no radiographic manifestation in the early stages, depriving the dentist of one of the most objective diagnostic tools.

Symptoms from a crack or fracture may be slight and present for *several months* before an accurate diagnosis is made,[5,16,19] which may be frustrating for both the patient and the dentist, and often causes the patient to develop a subsequent loss of trust and confidence in the dentist. The final diagnosis is typically reached at a relatively late stage of these conditions, often after complications have already

occurred. Complications may include a subsequent fracture of the tooth or cusp or significant periradicular bone loss associated with a VRF as seen by radiographic examination (see also Chapter 2). For this reason, this chapter emphasizes the *early diagnosis* of these conditions. The collection of signs and symptoms associated with cracks and fractures may be confusing; the clinician should consider these as indicators during the examination and diagnostic process.

Fracture Mechanics

Fracture mechanics is the field of biomechanics concerned with the propagation of *cracks* in a given material until the formation of the final catastrophic *fracture*.[7,53] Many terms have been loosely used in the dental literature to describe the clinical entities that are the subject of this chapter.[5,9,18,36] Here, the term *crack* will be used in the biomechanical sense: a *partial discontinuity* in a material that may propagate and eventually lead to a complete discontinuity, also known as a *fracture*.[7,53] In this context, a cusp may be described as *cracked*, a condition that may continue over time into a *fractured cusp*. A *cracked tooth* generally presents in a longitudinal direction, along the long axis of the root. It may be so called until a final fracture occurs that separates the tooth into two parts, a condition termed a *split tooth*. Similarly, microcracks may appear in the radicular dentin of an endodontically treated tooth, and these cracks may propagate with time until a *vertical root fracture* occurs, at which point the full thickness of the dentinal wall shows discontinuity: a through-and-through fracture. Such a fracture may be *incomplete*, involving one wall of the root, or *complete*, separating the root into two parts (Fig. 22.1).[81]

Cracked and Fractured Cusps

DEFINITION

A *cracked cusp* is characterized by a crack between a cusp and the rest of the tooth structure, to the extent of allowing microscopic flexure upon mastication. This crack typically does not involve the pulp. With time, the crack may propagate, eventually resulting in a *fractured cusp*.[5]



Fig. 22.1 Complete vertical root fracture. A cross section of a mesial root of a mandibular molar with a complete vertical root fracture. The fracture is in the buccolingual plane and extends from the buccal convexity of the root to its lingual convexity.

DIAGNOSIS

Patient History

In the case of a cracked cusp, the patient history is the most important tool for making a diagnosis. The patient will likely complain about pain when chewing, to the extent of not being able to chew on the side on which the crack occurred.[18,36,70] The patient will also often state that the condition existed for a relatively long period of time and that his or her dentist could not find the source or visualize any radiographic findings to suggest a cusp fracture.[3,16,48] When asked whether the pain is sharp or dull, patients will typically report a sharp pain that makes them immediately stop chewing on that side. The diagnostic challenge is attempting to determine which cusp of which tooth is involved, as patients often have difficulties determining the specific location of the discomfort.[16,18,36,70] Because the pain has a pulpal origin, the patient's proprioception may not be accurate, as no periodontal ligament is involved. Occasionally, the masticatory pain may radiate to nondental locations on the same side of the face,[16,48,72] as also described in Chapter 4.

Clinical Manifestation

Early Manifestation. The typical characteristic of a cracked cusp is generally a sharp pain upon chewing, although the affected tooth may not be sensitive, or only selectively sensitive, to percussion.[71] The tooth is vital, and its response to a cold stimulus may be normal; but with time, this response may resemble a reversible pulpitis, which may be either localized or referred to other odontogenic or nonodontogenic locations.[16,48,72]

Cracked cusps are often associated with extensive occlusal restorations,[3,29,72,79] which may undermine and weaken the cusp and predispose it to initiating or perpetuating a crack from occlusal forces. Nevertheless, a cracked cusp may also be present in intact nonrestored teeth or teeth that are minimally restored.[13,71]

Late Manifestation. With time, a crack under a cusp may propagate and result in a fractured cusp. If the fracture line occurs supragingivally, the fractured cusp will simply separate from the tooth. However, if the fracture line extends subgingivally, gingival fibers or the periodontal ligament will often retain the fractured cusp. Initially, it may be possible to move the cusp by wedging a sharp explorer into the fracture line, making the fractured cusp more visible. Often, from continued mastication, a localized and more acute pain may develop secondary to the movement of the fractured fragment on the underlying tooth structure. The pulpal pain that is typical at an earlier stage (the cracked cusp) will typically resolve once a complete fracture occurs. However, with the resulting tooth structure having exposed dentin (or even a pulp exposure), a resulting reversible or irreversible pulpitis may ensue.

Diagnosis

A cracked cusp may be diagnosed, to a large extent, on the basis of the patient history. To locate the affected tooth, a *biting test* should be performed using a Tooth Slooth (Professional Results, Laguna Niguel, CA) or a similar device (Fig. 22.2).[9,12] The device is composed of a small pyramid with a flattened top that is placed on a selective cusp, while the wider part of the



Fig. 22.2 The Tooth Slooth device. Application for a bite test: the tip of the pyramid is touching the tested cusp while the wide base is supported by multiple contacts.

device is applied to opposing teeth while the patient occludes. The application of these forces to a cracked cusp will generate a sharp pain, which may occur upon pressure or released.[3,35,44] The patient will typically state that this sensation has reproduced the sensation of their chief complaint.

Magnification with such devices as loupes or an operating microscope can be helpful when looking for a crack. If the tooth does not have an extensive intracoronal restoration, *transillumination* may also assist in revealing the crack line. If the tooth has a large restoration, the removal of the restoration may facilitate the effective use of this diagnostic tool (Fig. 22.3). The light source should be intense but with small dimensions; it is applied to the tooth at the area of the suspected cusp fracture, with the lights of the dental unit, microscope, and room extinguished. The light penetrates the tooth structure up to the crack, leaving the part beyond the crack relatively dark (see Fig. 22.3).

Once the crack propagates, resulting in a fractured cusp, the diagnosis becomes more objective, as the fractured cusp will either be missing or can be moved by wedging an explorer into the fracture line (Fig. 22.4).

ETIOLOGY

Extensive intracoronal restorations may be a predisposing factor for cracked and fractured cusps.[3,29,72,79] Otherwise, the etiology of these conditions is similar to that of cracked teeth (see Cracked and Split Teeth, Etiology).

TREATMENT PLANNING

Cracked Cusp

Treatment should consist of protecting the affected cusp from occlusal forces, both to prevent pain while chewing and to prevent the propagation of the crack into a full fracture. This can be facilitated with restorative treatment and an occlusal evaluation. Full-coverage crown or onlay is recommended,[5,16,40] although bonded composite restorations have also been proposed.[28,65] One should keep in mind that if the cracked cusp is not protected, the tooth may eventually fracture, complicating the treatment or even

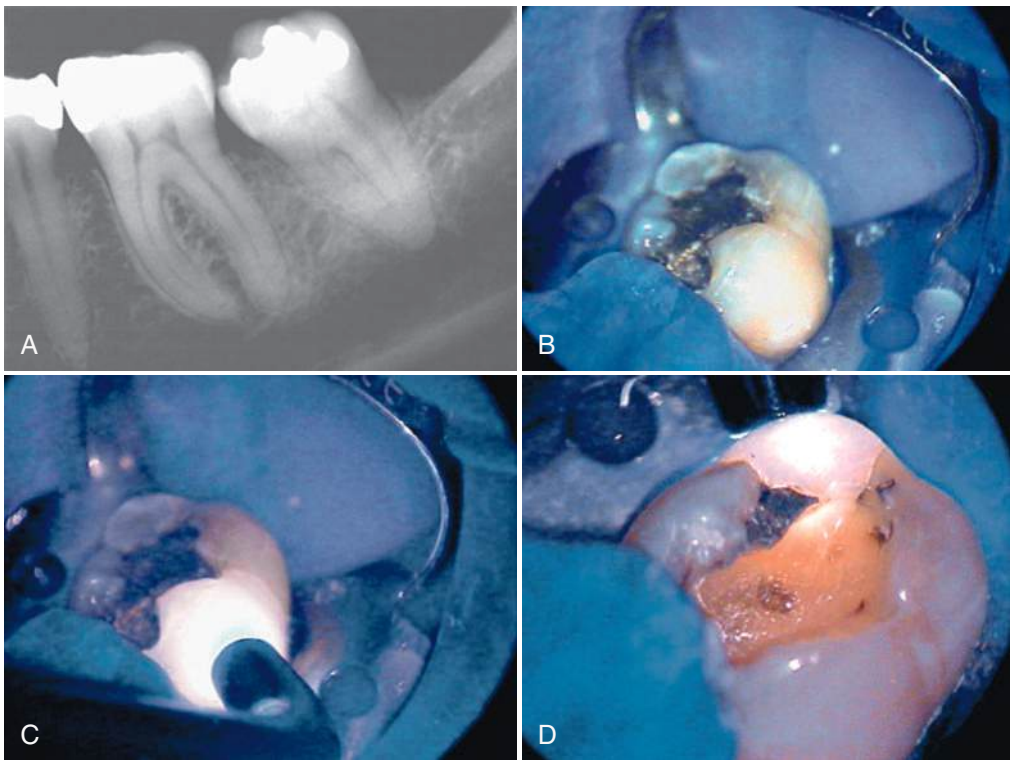


Fig. 22.3 Transillumination for the detection of a cracked cusp/tooth. An intense but small light source is applied to the suspect tooth, preferably in relative darkness. The light is transmitted through the tooth structure but is reflected from the crack plane, leaving the area behind the crack in darkness.

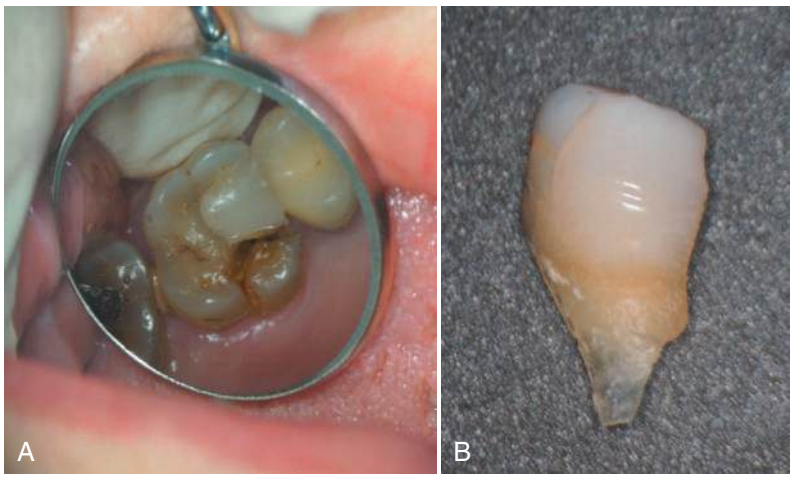


Fig. 22.4 A fractured cusp. **A,** The mesiopalatal cusp of a maxillary first right molar was fractured. The fractured cusp was movable and was retained by the periodontal ligament. **B,** The cusp was removed, and the tooth was considered restorable. It was treated with root canal treatment and a crown.

compromising the salvageability of the tooth. If the fracture plane extends apically into the root, the tooth may potentially become nonrestorable.[5,9,12] Endodontic treatment is indicated only if signs and symptoms of pulpal pathosis are observed. In addition, if the removal of the cracked cusp and associated restoration will result in little or no remaining coronal tooth structure, then elective root canal treatment may be necessary for prosthetic reasons. When such a treatment plan is selected, one should also perform an occlusal reduction of the tooth as soon as possible to remove the tooth from active occlusion. The patient should be instructed to be careful when chewing until the tooth is restored with a crown.

Fractured Cusp

The treatment of a fractured cusp depends on the amount of tooth structure remaining. If the missing cusp is limited in size, then the conservative restoration of a bonded composite resin may be indicated to cover the exposed dentin. In contrast, when a larger fragment has fractured and is either removed or missing, a full crown or an onlay may be necessary.

In certain cases, when cracked cusps are found in intact teeth or in teeth with no extensive restoration, it is difficult to predict the direction in which the crack is propagating. Therefore, in these cases, when considering endodontic and restorative treatment, the patient should be advised as to the potential decrease in prognosis, as described later.

Cracked and Split Teeth

DEFINITION

As previously described, a *cracked tooth* has "a thin surface disruption of enamel and dentin, and possibly cementum, of unknown depth or extension." It exhibits a crack that incompletely separates the tooth crown into two parts. If the crack is allowed to propagate longitudinally, the tooth will eventually fracture into two fragments, resulting in a *split tooth*. The American Association of Endodontists describes a split tooth as "a continuation of a crack or vertical root fracture whereby the fractured segments are completely separated longitudinally; it may occur buccal-lingually or mesial-distally; it may cause an isolated periodontal defect(s) or sinus tract; it may be radiographically evident."

DIAGNOSIS

Patient History

In cases of a cracked tooth, the patient history may be similar to that of a cracked cusp; there may be a sharp pain upon mastication and prolonged inability of the dentist to diagnose the source of the pain.[16,19] Similar to a cracked cusp, the diagnosis of a cracked tooth is often made on the basis of the patient history alone. Often it is challenging for the practitioner to determine the location of the offending tooth. With time, the patient may report that he or she *used to have* a sharp pain and now experiences great sensitivity to cold stimuli; the patient may even report, at a later stage, that the pain has subsided. These observations are consistent with pulpitis or pulp necrosis, which may develop in the affected tooth over time.[13]

Clinical Manifestation

Early Manifestation. Cracked teeth may have extensive restorations with a weakened crown, or they may have minimal or no restorations. A cracked tooth begins with a crack in the clinical crown, which may gradually propagate in an apical direction.[3,5,28,48] Such cracks typically run in the mesiodistal direction, often splitting the crown into the buccal and lingual fragments. In the early stages, the tooth may be vital and painful to mastication. The pain may be sharp, to the extent that the patient is unable to chew on the affected side. This condition may persist for an extended period of time.[16,18,19] The pain may be localized or referred to any tooth, maxillary or mandibular, on the same side of the mouth.[16,48,72] No radiographic manifestations are present at these early stages, as the crack is microscopic and runs perpendicular to the x-ray beam. The affected tooth may or may not be sensitive to percussion at this point, and pulp testing may be normal or indicative of increased sensitivity to cold stimuli.

Late Manifestation. The late manifestation of a cracked tooth may include pulp involvement and potentially the loss of pulp vitality[13] or apical propagation of the fracture resulting in a split tooth. In one investigation,[13] 27 nonvital molar and premolar teeth that had minimal or no restorations or caries were studied. Upon extraction, these teeth were examined under a surgical operating microscope or using micro–computed tomography (CT) analysis. In all of these teeth, a longitudinal crack was observed, which extended to the pulp and often to an external surface of the root. Although the study had a limited sample size, the clinician should understand the potentially poor prognosis from cracked teeth, especially when the crack is suspected to be the *cause* of pulp necrosis ("fracture necrosis") (Fig. 22.5).[13]

Pulp involvement occurs more often in cases of centrally located cracks than in cracks with a more buccal or lingual location (i.e., extending from marginal ridge to marginal ridge through the central fossa).[18,91] These centrally located cracks commonly affect the roof of the pulp chamber at a later stage. Consequently, pulp vitality may be compromised and later lost due to bacterial penetration through the crack. The sharp pain upon mastication that is typical of the early stage may disappear once pulp vitality is lost. Consequently, apical periodontitis in an apparently intact molar may be a late manifestation of an untreated case of a cracked tooth.[13] When pulp necrosis occurs, the radiographic manifestation may be an apical radiolucency, which is undistinguishable from that of apical periodontitis (see Fig. 22.5).

A crack may propagate over time through the pulp chamber and into the root, resulting in a complete fracture that separates the tooth into two parts, a condition termed *split tooth*. This is defined in the AAE *Glossary of Endodontic Terms* as a longitudinal fracture ("a root fracture extending in the axial plane within the tooth") that becomes "a continuation of a crack or vertical root fracture whereby the fractured segments are completely separated longitudinally."[3a] When this split occurs, the resulting parts of the tooth may be movable by wedging a sharp explorer into the fissure.[5,9] Given time and masticatory forces, more evident movement of the parts may be observed. The radiographic presentation at such a late stage may eventually develop into a diffuse radiolucency surrounding the root. At this late stage, narrow isolated deep periodontal pockets may be present.[9] However, such pockets are typically located mesially or distally and, if adjacent teeth are present, they will be difficult if not impossible to detect.

Cracked and split teeth may present with a large and variable collection of signs and symptoms that are potentially confusing.[16,18,79] Only by being aware of the *process* leading from early to late manifestations can clinicians interpret these signs and symptoms and identify the specific point they are encountering on the potential timeline of this process.

A *definitive* combination of factors, signs, and symptoms that when collectively observed allows the clinician to conclude the existence of a specific disease state is termed a *syndrome*. However, given the multitude of signs and symptoms that cracked roots can present with, it is often difficult to achieve an objective definitive diagnosis. For this reason, the terminology of *cracked tooth syndrome*[18] should be avoided.

Diagnosis

Early detection of a cracked or split tooth is imperative to resolve the patient's symptoms as well as increasing the prognosis. However, the use of the Tooth Slooth device (see Fig. 22.2) may or may not provide as clear a result for a symmetrically cracked tooth, as each of the parts of the tooth may be rather stable. Asking the patient to chew on a cotton roll[9] or on the tip of a cotton-tip applicator placed at

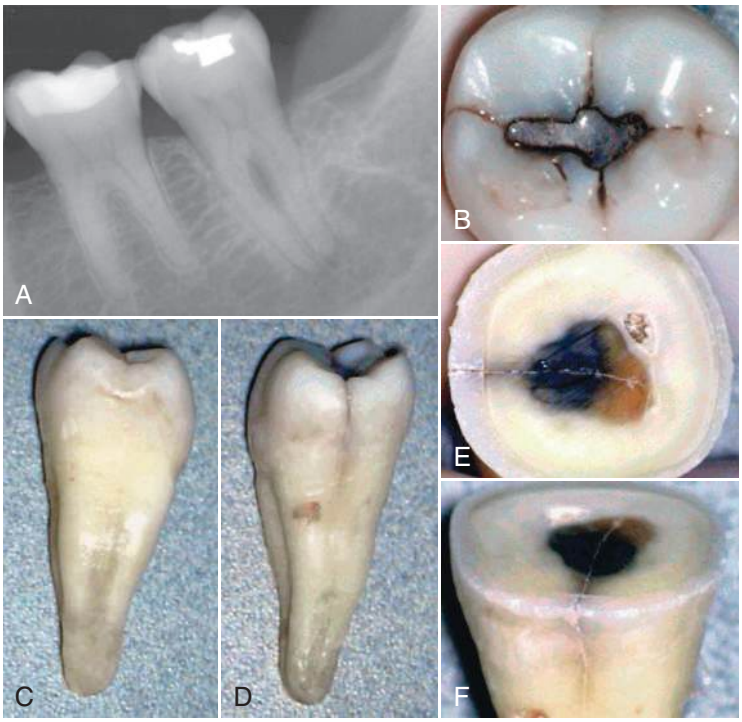


Fig. 22.5 A case of fracture-induced necrosis. A tooth with minimal or no restoration or caries is unlikely to become nonvital. **A,** This radiograph of a mandibular second molar shows a restoration that is distant from the pulp chamber, yet the tooth is nonvital and symptomatic. **B,** On occlusal examination, a slight crack is observed on the distal marginal ridge. **C,** After extraction, the mesial aspect of the crown and root shows no indication of a fracture. **D,** However, the distal aspect of the crown and coronal root shows the fracture. **E** and **F,** When the crown is sectioned, the crack can be observed to extend well into the pulp chamber.

a particular site may reproduce the symptoms. Nevertheless, this method may not indicate whether the source is a maxillary or mandibular tooth and may need further measures to pinpoint the involved tooth. *Magnification* using either loupes or an operating microscope can be helpful for detecting a fracture line. In addition, dyes, such as methylene blue or tincture of iodine, which are applied either to the outer surface of the crown or to the dentin after the removal of an existing intracoronal restoration, can be helpful for visualizing the crack (see Fig. 22.3). Transillumination can also be applied to the suspected tooth and, if the tooth has no restorations, this method may yield an impressively straightforward diagnosis (see Fig. 22.3). Anesthetizing the suspected tooth, followed by asking the patient to chew again on the cotton roll, may further confirm the diagnosis and finally differentiate the origin as a mandibular or maxillary tooth. At a later stage, when splitting of the tooth has occurred, wedging of a sharp explorer into the fracture line will provide a clear diagnosis of a split tooth.

In general, the diagnosis of a crack in a tooth can be difficult. When there are signs and symptoms of a pulpitis or necrosis, it is incumbent upon the clinician to determine the source that initiated the signs and symptoms. In the case of a problematic tooth with no apparent reason for a pulpitis or necrosis, like a tooth with minimal or no caries, restoration, or trauma, a crack or fracture must be considered.[13] In some cases, an objective diagnosis may not be possible; however, with the possibility of a crack or fracture, the patient should be advised of a potential decrease in the endodontic or restorative prognosis.[5,12]

ETIOLOGY

Masticatory forces are the cause of cracked teeth.[22,35,46,48,71,72,91] Thus, the dietary habit of chewing on coarse food has been proposed as a contributing factor.[22,91] Bruxism or clenching of the teeth as well as occlusal prematurities are also frequent causes of cracked te eth.[18,22,35,46,48,71,72,91] For this reason, certain teeth may be more prone to developing cracks, such as mandibular second molars and maxillary premolars.[18] Masticatory habits such as chewing ice may also predispose teeth to cracks. The term *fatigue root fracture*, proposed by Yeh,[91] encompasses all of these causes.

In certain cases, traumatic injuries, such as a severe upward blow to the mandible (e.g., during a car or sports accident) can also cause a tooth crack or fracture. Another potential cause is the unexpected chewing of a hard object (e.g., a cherry pit or an un-popped corn kernel in popcorn). The occlusal forces applied by the first molars are as high as 90 kg,[46] which, when fully applied to a small surface area, may damage the tooth structure. However, in most cases, a tooth crack can be attributed to no specific cause other than normal or excessive masticatory forces.[22,35,46,48,71,72,91]

TREATMENT PLANNING

Cracked Tooth

When a cracked tooth is either suspected or determined, the patient should be informed that the prognosis is reduced and is sometimes questionable.[5,9,12] Protecting a tooth from the propagation of a crack and improving comfort while chewing are the principal goals in the treatment of cracked teeth. Both goals can often be immediately achieved by placing an orthodontic band around the tooth[3,9,16] or by placing a provisional crown. These procedures allow the clinician to evaluate the extent of pulp involvement by checking whether the pulpal symptoms subside in response to the intervention.[3,9,16]

Protecting the tooth from further splitting forces by the placement of a permanent crown is essential in these cases.[3,5,9,16,40] Unfortunately, a crown alone is often not sufficient to resolve symptoms, and endodontic treatment may be considered prior to the placement of the permanent crown, depending on the pulpal symptoms.[9]

A root canal treatment followed by a permanent crown has the benefit of immediately eliminating the long-lasting, painful symptoms as well as early protection from occlusal forces that may cause propagation of a cracked tooth to become a split tooth. Nevertheless, when 127 cracked teeth with reversible pulpitis were treated with a crown alone, 20% of these teeth converted to irreversible pulpitis within 6 months and required root canal treatment.[52] In contrast, none of the other teeth needed a root canal over a 6-year evaluation period.[52] However, it should also be understood that varying percentages of crowned teeth, cracked or noncracked, may also require endodontic treatment, merely from the trauma of the crown preparation. By comparison, it has been shown that when suspected cracked teeth were restored with resin bonded restorations, only 7% required subsequent endodontic treatment or extraction.[65]

After removal of an intracoronal restoration or when penetrating the dentin to prepare for endodontic access, one may observe discoloration along the crack in the dentin. Once the access cavity has been completed, the pulp chamber floor and the distal and mesial walls of the pulp chamber should be carefully inspected to check for the presence and extent of any cracks. Upon evaluating 245 restored teeth, cracks were observed preoperatively in 23.3% of the teeth; however, when the restorations were removed, 60% of these teeth were found to have cracks.[1] Methylene blue dye may be helpful in this type of examination. If a crack is found that reaches from the mesial wall, through the floor of the pulp chamber and into the distal wall, then the prognosis for the tooth is poor, and extraction should be considered (see Fig. 22.5).[5,9,12] If the crack does not reach the pulp chamber or is limited to the coronal parts of the mesial or distal wall, the subsequent protection of the tooth with a crown may save the tooth. Nevertheless, as mentioned previously, the patient should be advised that the treatment success may be compromised and that long-term follow-up will be required.[5,9,12]

Given that such cracks occasionally occur in teeth with minimal or no restorations and that all of the pain may cease once the pulp is removed, it may be tempting to limit the final restoration to an intracoronal amalgam or composite restoration. This temptation should be resisted by all means, as the forces that caused the crack are still present, and the apical propagation of the crack and loss of the tooth is still likely.

Split Tooth

When the tooth is split either through its entire length or obliquely (Fig. 22.6), extraction is typically the only treatment

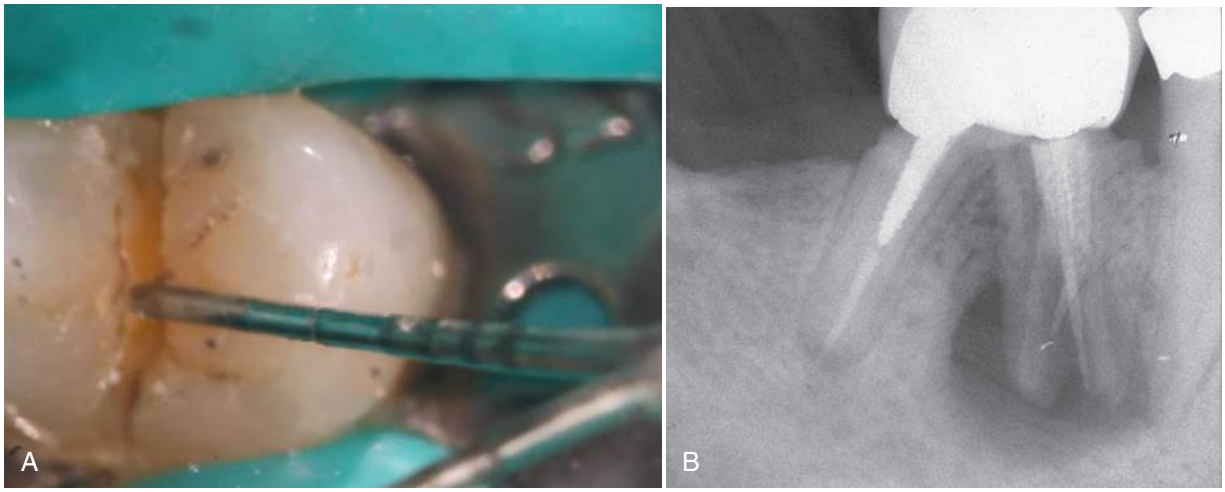


Fig. 22.6 A, Observing split tooth by probing or pushing cusp. **B,** Radiographic evidence of split tooth. (**A,** Courtesy Dr. M. Malek.)

option.[5,9] However, if the fracture line is such that the split results in large and small segments, and if the removal of the small fragment preserves enough tooth structure that is restorable, then retention and restoration of the tooth may be considered.[5] If this is a consideration, the patient should be aware that periodontal crown lengthening may be necessary. Proper interdisciplinary management of cases like this is imperative so as to ensure that the prognosis is evaluated and maximized.

Vertical Root Fracture

DEFINITION

A *vertical root fracture* is defined in the AAE *Glossary of Endodontic Terms* as a longitudinal fracture "in the root whereby the fractured segments are incompletely separated; it may occur buccal-lingually or mesial-distally; it may cause an isolated periodontal defect(s) or sinus tract; it may be radiographically evident"[3a] (Figs. 22.7–22.9; see also Fig. 22.1). Previous definitions described VRFs as initiating from the apical root and traveling coronally, or only as a result of endodontic treatment. In the latest edition of the AAE *Glossary of Endodontic Terms*, these stipulations have been removed. It is differentiated from a split root in that segments associated with the fracture are not completely separated.

DIAGNOSIS

Patient History

In the case of a VRF, a patient may complain of pain or sensitivity related to or adjacent to a specific tooth. Sensitivity and discomfort while chewing are also common complaints.[81] Swelling may occasionally occur in the area. There is often a long history of failing to diagnose the cause of the pain and discomfort. A history of repeated clinical and radiographic examinations that revealed no cause for the pain is also common in VRF cases. After recent endodontic retreatment, if the symptoms remain and the dentist is unable to determine the cause of the symptoms, the patient may lose confidence in the dentist. Often retreatment or surgical retreatment may have been attempted to reveal the accurate diagnosis.[1] Unfortunately, such ineffective treatment attempts may only worsen the dentist-patient relationship.

Clinical Manifestations

Susceptible Teeth and Vertical Root Fracture Location. VRFs are commonly associated with endodontically treated teeth with or without a post.[5] Nevertheless, VRFs can also occur in teeth with no previous root canal treatment.[22] The most susceptible sites and tooth groups are the maxillary and mandibular premolars, mesial roots of the mandibular molars, mesio-buccal roots of the maxillary molars, and mandibular incisors.[82] However, VRFs may occasionally occur in other teeth and roots as well. VRFs may progress in the bucco-lingual direction in these teeth and roots, which are typically narrow mesiodistally and wide buccolingually.[41] However, VRFs may also propagate diagonally, thus affecting the mesial or distal aspect of the root (see Fig. 22.7). VRFs may be initiated at any root level.[5] They may also be initiated at the apical part of the root and propagate coronally (see Fig. 22.8, *C*). Some VRFs originate at the coronal-cervical part of the root and extend apically (see Fig. 22.8, *A*), and in other cases a VRF may be initiated as a midroot fracture (see Figs. 22.8, *B* and 22.9, *C* and *D*).

It is commonly believed that VRFs begin as microcracks in the area of the root canal surface of the radicular dentin and gradually propagate outward until the full thickness of the radicular dentin is fractured.[5,10,21,57,88] Other studies[14,17,51,77,92] have shown that microcracks can also be initiated at the outer surface of the root and propagate inward. Therefore, the correlation between microcracks in the radicular dentin and the formation of VRFs should be further investigated.

Early Manifestation. In the early stages of a VRF, there may be pain or discomfort on the affected side of the tooth. In particular, the tooth may feel uncomfortable and sensitive upon chewing, although this pain is often of a dull nature, as opposed to the sharp pain typical of a cracked cusp or tooth with a vital pulp. As the fracture and subsequent



Fig. 22.7 Diagonal vertical root fracture (VRF) in a molar. A lingual view of a right first mandibular molar revealing a diagonal VRF in the mesial root.

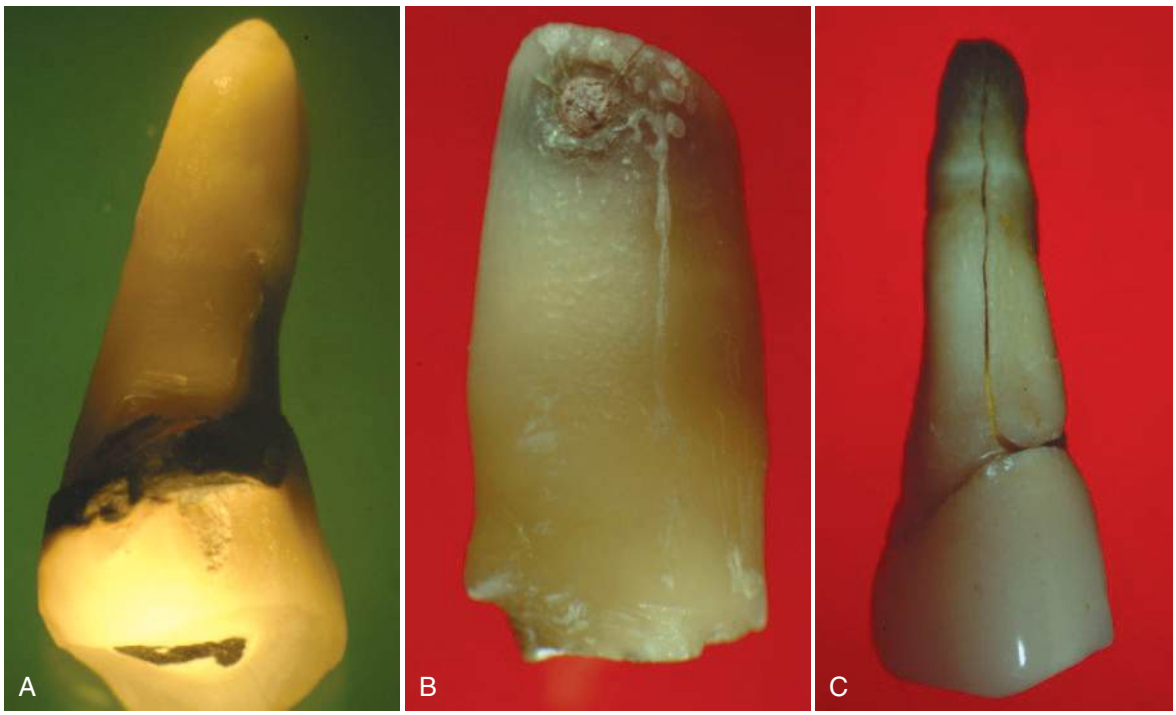


Fig. 22.8 Three types of vertical root fractures (VRFs). **A,** A coronally located VRF extending apically as far as one third of the root. **B,** A midroot VRF extending along the middle third of the root. **C,** An apically located VRF extending coronally as far as the apical two thirds of the root.

infection progresses, swelling often occurs, and a sinus tract may be present. Noteworthy is that the location of a sinus tract associated with a VRF is more coronal than a sinus tract that is typically associated with a case of a chronic apical abscess (Fig. 22.10).[80,82] These signs and symptoms are frequently similar to those encountered from nonhealing root canal treatment.[58,87] In the early stages, radiographic findings are unlikely because (1) the root canal filling may obstruct the detection of the fracture (Fig. 22.11, *A*), and (2) the bone destruction (which still has limited mesiodistal dimensions) may be obstructed by the superimposed root structure (Fig. 22.12).

Often in a case of a VRF, there is the presence of a deep, narrow, and isolated periodontal pocket. This presentation may be inconsistent with the surrounding periodontal status.[80,82,86] This specific type of periodontal defect occurs

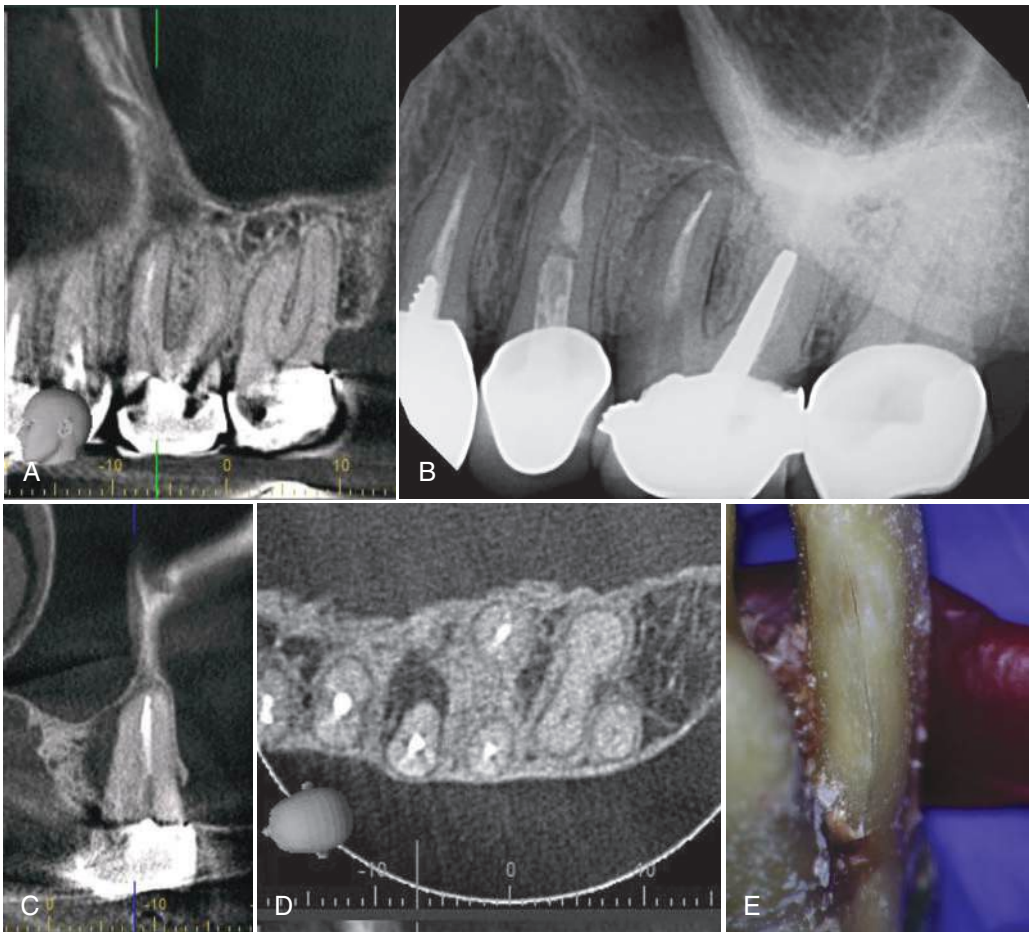


Fig. 22.9 **A** to **E,** Vertical root fractures observed with periapical radiograph, cone-beam computed tomography and clinically after extraction. (Courtesy Dr. R. Ganik.)

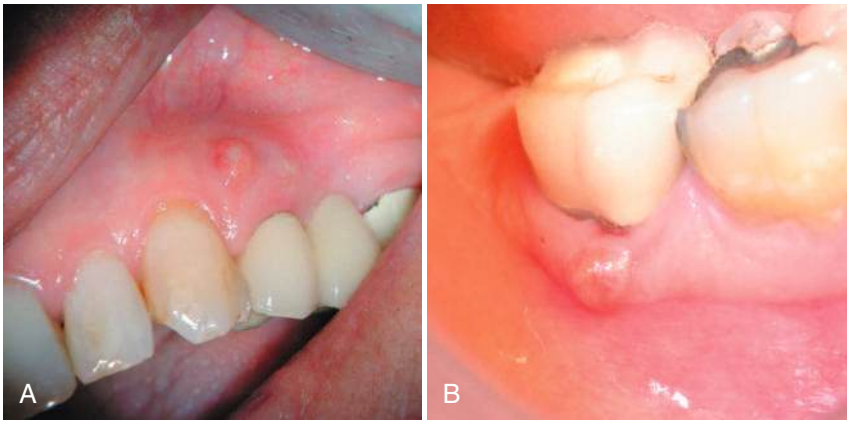


Fig. 22.10 Coronally located sinus tracts. **A,** A draining sinus tract at a coronal location originating from a buccal vertical root fracture (VRF) in the first left maxillary premolar. **B,** A draining sinus tract at the gingival margin of a right first mandibular molar with a buccal VRF in the mesial root.

secondary to the bony dehiscence caused by the VRF and is substantially different in its presentation from the pockets caused by advanced periodontitis (discussed later).

Late Manifestation. A longstanding VRF is easier to detect. The major destruction of the alveolar bone adjacent to the root has already occurred, allowing the VRF to be more likely revealed in a periapical radiograph (see Fig. 22.6, *B*). One of the most typical radiographic signs is a *J-shaped* or *halo radiolucency*, which is a confluence of periapical and periradicular bone loss (i.e., bone loss apically combining with the bone loss on the side of the root, extending coronally) (Fig. 22.13).[83,84] In addition, the pocket now approximating the fracture, which was initially tight and

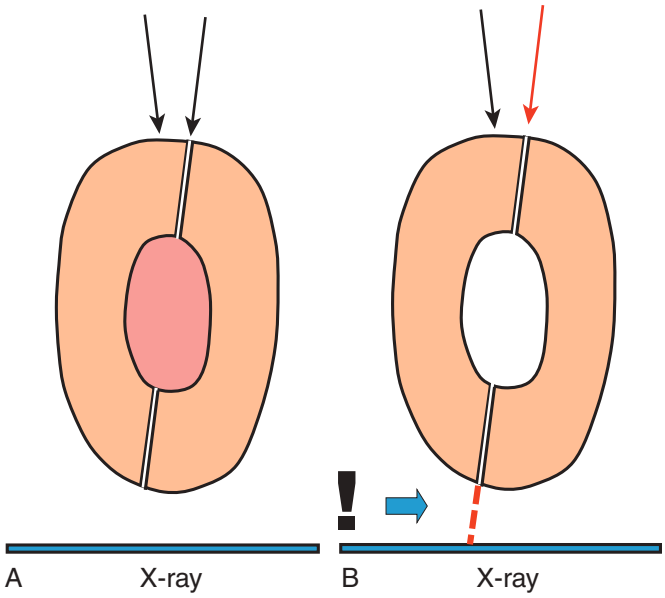


Fig. 22.11 Radiographic examination of filled versus empty canals. **A,** The buccolingual projection of a filled root will fail to detect a vertical root fracture (VRF) at an early stage. **B,** The removal of the root filling and use of radiography at different mesiodistal angulations may reveal the VRF.

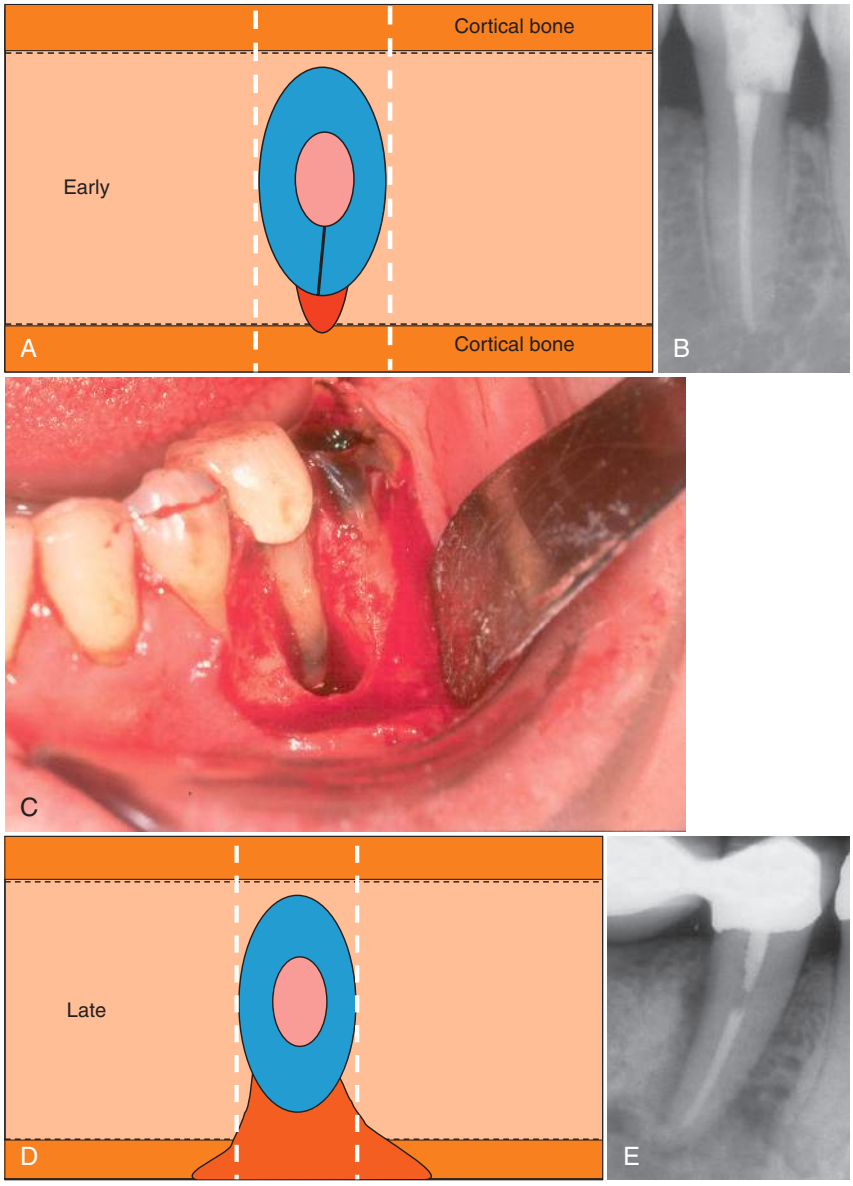


Fig. 22.12 Early versus late radiographic presentation of a vertical root fracture–associated bone defect. At an early stage, a bone defect *(red)* is not likely to be detected in a periapical radiograph, as the root will overlap with the defect **(A** and **B).** At later stages, when major damage has occurred to the cortical plate **(C),** the bone defect may be large enough to extend beyond the silhouette of the root **(C** and **D)** and appear as a radiolucent defect along the root **(E).** (Surgical image courtesy Dr. Devora Schwartz-Arad, Ramat-Hasharon, Israel.)

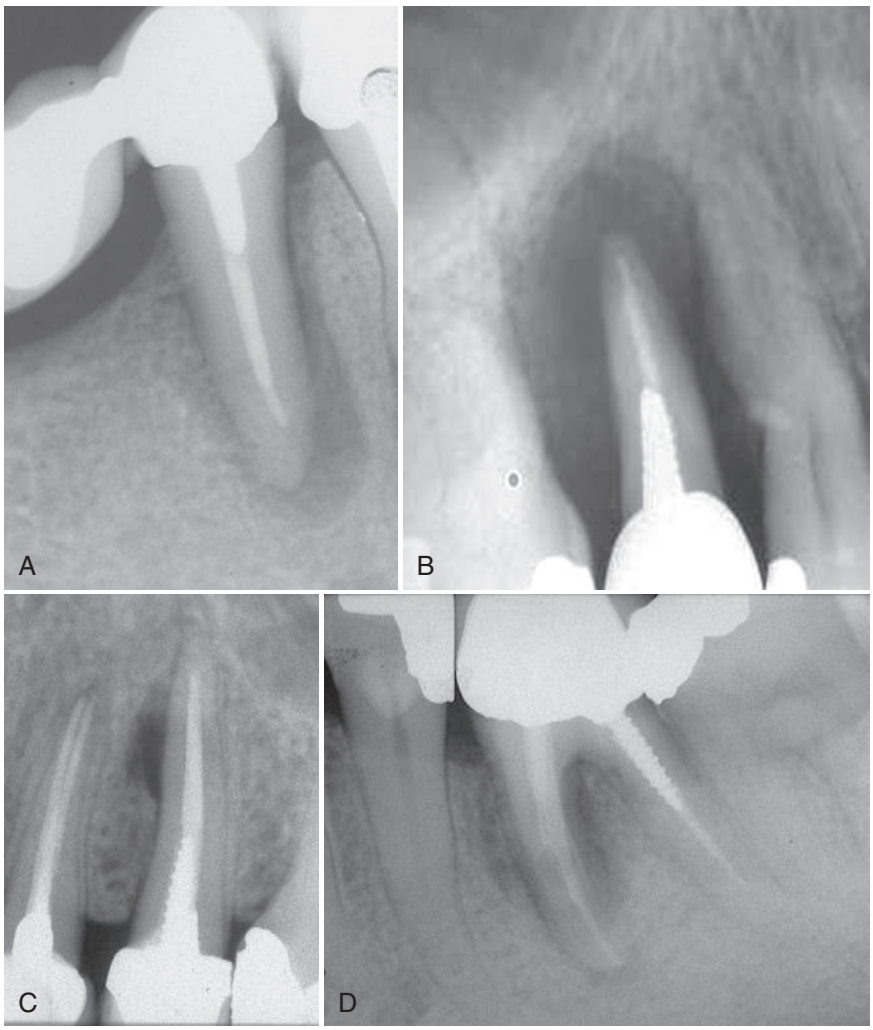


Fig. 22.13 Radiographic presentations of longstanding vertical root fractures (VRFs). **A,** J-shaped "halo" associated with a VRF in a second right mandibular premolar. **B,** Extensive bone damage associated with a complete VRF in a second right maxillary premolar. **C,** Limited bone damage associated with a midroot VRF in a second left maxillary premolar. **D,** Bone damage associated with a VRF in the mesial root of a first left mandibular molar. All such presentations are typical of longstanding VRFs.

narrow, may become wider and easier to detect. In longstanding cases in which the bone destruction is extensive, the VRF may result in a split root whereby the segments of the root separate, resulting in radiographic evidence clearly revealing an objective split root (Fig. 22.14).

Diagnosis

Importance of Early Diagnosis. Accurate and timely diagnosis is crucial in VRF cases, allowing the extraction of the tooth or root before extensive damage to the alveolar bone occurs. Early diagnosis is particularly important when implants are a potential part of the future restorative process; when an extraction is performed at an early stage, the uncomplicated placement of an implant is more likely. When the tooth is extracted after extensive damage has already occurred, bone regeneration procedures may be required,[43] adding additional cost and time to the restoration process.

The American Association of Endodontists stated in 2008[5] that a sinus tract and a narrow, isolated periodontal probing defect associated with a tooth that has undergone a root canal treatment, with or without post placement, can be considered pathognomonic for the presence of a VRF.

However, the combination of the following two factors makes the early diagnosis difficult: (1) many of the clinical symptoms associated with VRFs mimic apical periodontitis or periodontal disease, and (2) the narrow and tight pocket associated with early stages of VRF is difficult to detect using rigid probes (see the Vertical Root Fracture Pockets section), especially when the pocket is interproximal. Consequently, a delay in the accurate diagnosis—or a misdiagnosis of a VRF—may often occur.

Misdiagnosis of Vertical Root Fractures. Certain cases of longstanding VRFs are so discernible that no dentist should miss the diagnosis (see Fig. 22.14). Nevertheless, two retrospective case series, one by Fuss et al.[33] and another by Chan et al.,[22] reported that general practitioners often misdiagnose VRFs. The teeth that were extracted in these studies had often been diagnosed as endodontic failures or refractive periodontal pockets, only to realize after extraction that in some of them the actual cause was a VRF.

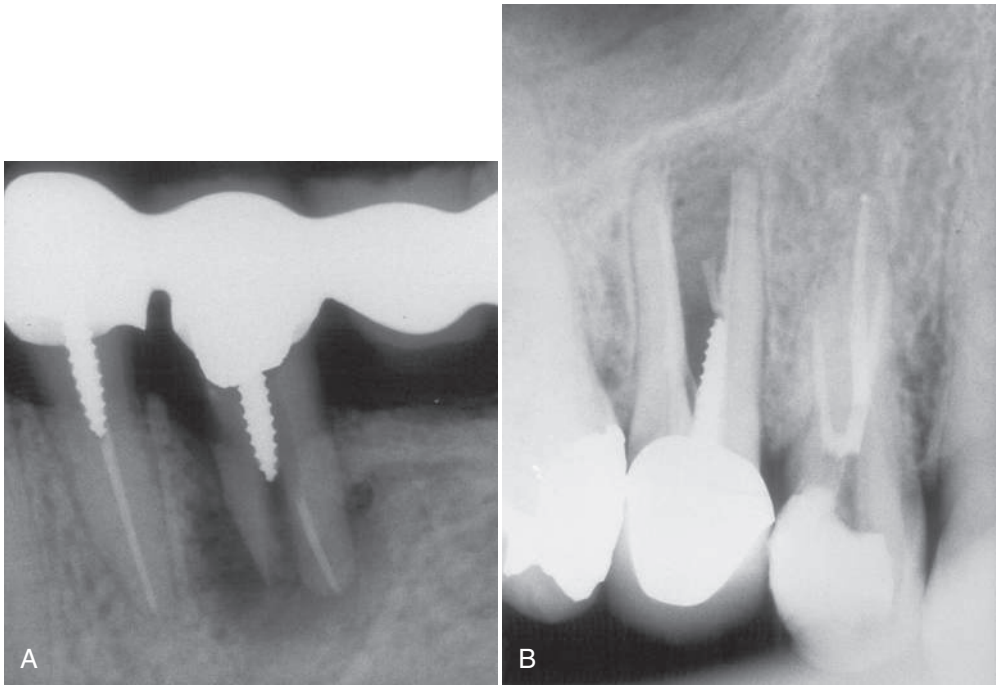


Fig. 22.14 Radiographs of longstanding vertical root fractures. No special diagnostic skills are needed to diagnose such cases.

Vertical Root Fracture Pockets. The pockets that are typical of the early stages of VRFs differ substantially from the deep pockets associated with advanced periodontal disease. The deep pockets associated with periodontal disease develop as a result of the bacterial biofilm that initially accumulates at the cervical areas of the tooth and the destructive host response to these bacteria.[39] Therefore, deep periodontal pockets are typically wider coronally and easier to probe. This pocket anatomy allows an easy insertion of a rigid periodontal probe (Fig. 22.15). These types of periodontal pockets typically present with the deeper part of the pocket at the mesial or distal aspects of the tooth. In addition, periodontal disease often affects groups of teeth rather than an isolated location of a single tooth.

The pockets associated with VRFs develop due to bacterial penetration into the fracture, triggering a destructive host response that occurs in the periodontal ligament along the entire length of the fracture. These bacteria may

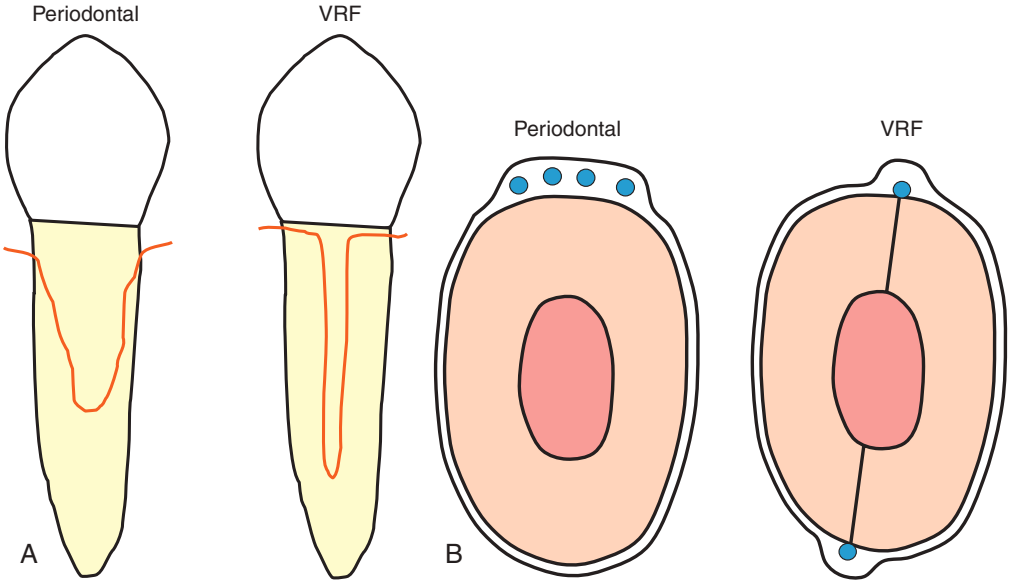


Fig. 22.15 Vertical root fracture *(VRF)* pocket. **A,** Periodontal pockets *(left)* are wide coronally, whereas VRF pockets *(right)* are narrow and deep. **B,** Periodontal pockets *(left)* are loose and allow probing at various sites, whereas VRF pockets *(right)* are narrow and tight. If not checked carefully at every millimeter of the sulcus, an early VRF pocket can easily be missed. Note that periodontal pockets appear more commonly in the proximal sides of the root, whereas VRF pockets are more common on the buccal or lingual sides.

leak from an infected root canal[88]; however, when the VRF extends to the cervically exposed root, the microbes in the fracture may also originate from the oral cavity. In the early stages, the periodontal ligament is affected and destroyed along the longitudinal opening of the fracture, initially with a limited resorption to the adjacent bone. This permits the penetration of a periodontal probe. The pocket associated with a VRF is typically isolated and present only in a limited area adjacent to the affected tooth. This pocket is often located at the buccal or lingual convexity of the tooth. In the early stages, the pocket is deep but has a narrow coronal opening (see Fig. 22.15). The insertion of a probe first requires the detection of the coronal opening; often, light pressure is necessary for the insertion of the probe. Because the pocket is narrow, probe insertion may result in the blanching of the surrounding tissue (Fig. 22.16). This is specially the case when a more flexible plastic probe is used, as its coronal part is thicker than an equivalent metal probe (Fig. 22.17; also see Fig. 22.16). Therefore, the pocket associated with the early stages of a VRF is quite different from a typical periodontal pocket. This difference has been widely recognized, and terms such as *osseous defect*[28] and *probing defect*[5] have been used to emphasize the point. Nevertheless, these pockets do possess enough unique features to justify them being specifically termed a *VRF pocket*.

Rigid metal periodontal probes may be ineffective in probing VRF pockets in the early stages of a VRF. Given that the pocket is deep, narrow, and tight, the bulge and obstruction of the tooth's crown may prevent the insertion of a metal probe into the pocket (see Fig. 22.17). A flexible probe should be used instead, such as a probe available from Premier Dental Products (Plymouth Meeting, PA) (see Fig. 22.17) or a similar device. This type of flexible probe should be included on every endodontic examination tray and is an essential tool when checking for potential VRF pockets.

Tamse et al.[82] reported the presence of a typical VRF pocket in 67% of the VRF cases reported. However, because the early detection of such pockets is technique sensitive and because traditional metal probes were used in the aforementioned study, the incidence of these pockets may in fact be higher than reported. When a typical VRF pocket is present on the convex surfaces of the root, on the buccal or lingual aspects, it is likely that the root has a VRF. In contrast, when such a pocket is located at the furcation of a molar, the pocket may indicate either a VRF or a sinus tract from an apical abscess that drained at a point of least resistance at the furcation area. In cases with a furcation pocket, when a VRF diagnosis cannot be conclusively determined, a positive healing response to the elimination of infection (by periodontal therapy or initiating root canal treatment if necessary) may differentiate between these two types of pathoses.

Coronally Located Sinus Tract. Sinus tracts that originate from a chronic apical abscess are typically detected at the site of least bone resistance, against the apical part of the root or in the area of the junction of the attached gingiva and the oral mucosa. Sinus tracts that are associated with a VRF pocket are often found in a more coronal position, as the source is not from a periapical lesion (see Fig. 22.10).[81]

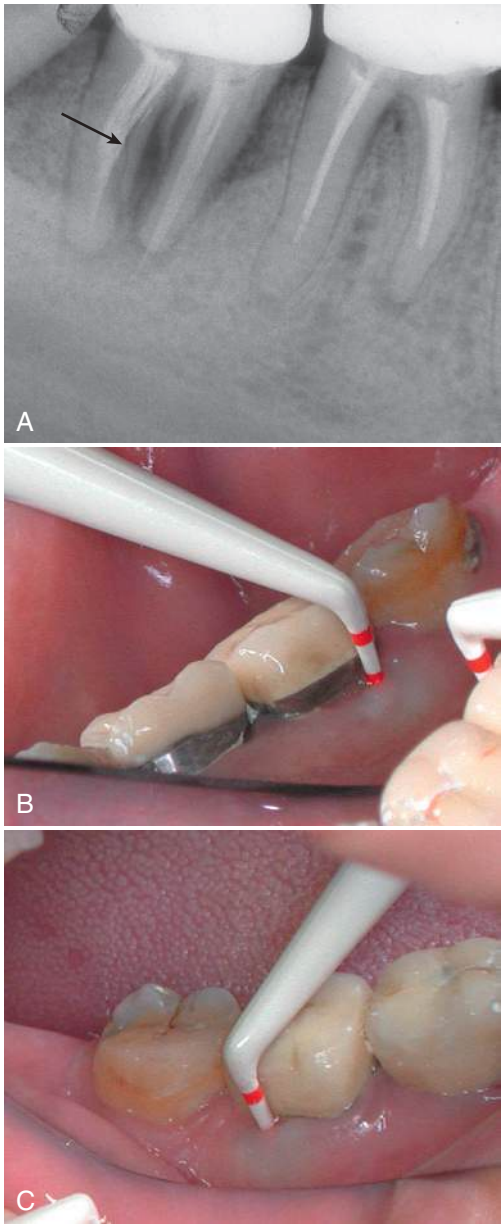


Fig. 22.16 Tight vertical root fracture (VRF) pockets. **A,** The distal root of a second right mandibular molar with a VRF. VRF pockets were found on both the lingual **(B)** and the buccal **(C)** sides. VRF pockets are tight, so inserting a probe into these pockets causes pressure blanching of the surrounding tissues.

In four clinical retrospective case series, coronally located sinus tracts were found in 13% to 35% of these cases.[59,80,82,86] As in the case of a VRF pocket, if the sinus tract is located at the furcation of a molar, this observation does not necessarily indicate a VRF, as periapical abscesses from reinfected root canal treatment can also drain in this coronal location.

Radiographic Features. A definitive VRF diagnosis can sometimes be made based on the radiographic appearance of a thin radiolucent line extending longitudinally down the root.[73] Such lines, however, are difficult to detect and are commonly not seen in routine periapical radiographs because either the root canal filling has "masked" the fracture

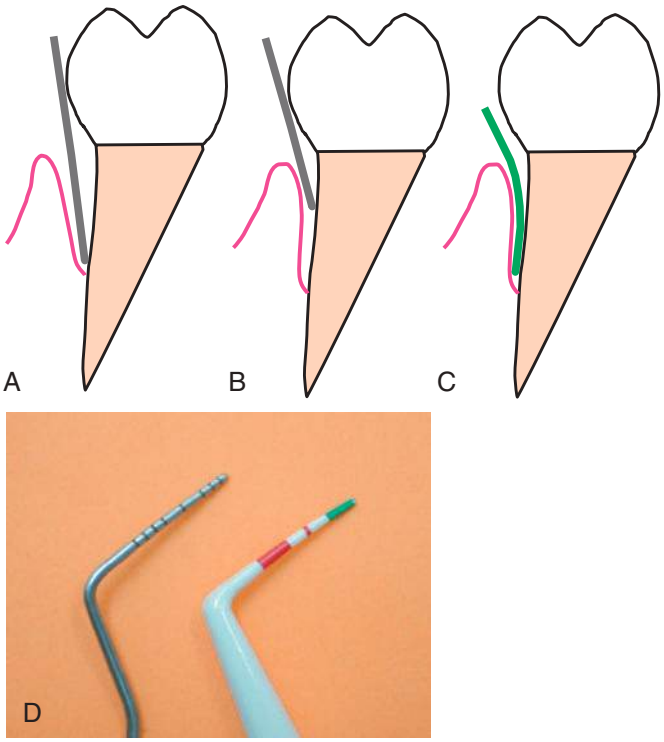


Fig. 22.17 Rigid versus flexible probes. **A,** In a loose periodontal pocket, a rigid metal probe can easily reach the depth of the pocket. **B,** In a tight, early-stage vertical root fracture (VRF) pocket, a rigid probe may be of limited value, as the bulge of the crown often prevents the insertion of the probe into the tight, deep pocket. **C,** A flexible probe **(D)** is more likely to detect VRF pockets at an early stage.

line or the angulation of the x-ray beam is not optimal for discerning the fracture (Fig. 22.18; also see Fig. 22.11). Rud and Omnell[73] claimed that it was possible to observe fracture lines in 35.7% of cases, but many of these cases were not true VRF instances. In clinical practice, it is still rare to observe a VRF on a radiograph, especially when only a single periapical radiograph is exposed. Such an observation requires the x-ray beam to align with the plane of the fracture as well as the fracture line not being superimposed over the radiopaque root filling. Therefore, two or three periapical radiographs should be exposed from different horizontal angulations when a fracture is suspected (Fig. 22.19; also see Fig. 22.18).[81] In most VRF cases, the clinician must make subjective *interpretations* or *predictions* based on the various patterns of periradicular bone destruction which, unfortunately, are also shared by other periodontal and endodontic-like lesions.[59,76,89]

In the early stages of a VRF, no radiolucent bone lesions may be observed[20] (see Fig. 22.12), which may be the reason why VRFs often remain undetected, delaying diagnosis and treatment. Rud and Omnell[73] correlated the direction of the fracture, the degree of bone destruction, and the radiographic appearance, and emphasized that the extent of bone destruction around a fractured root depends on the location of the root fracture and the time elapsed since the inception of the fracture. The significance of time was confirmed by Meister et al.,[59] who demonstrated that immediate radiographic detection is difficult due to the time required for bone resorption to occur or for the fractured segments to separate and be radiographically visible. In a study of the patterns of bone resorption in 110 VRF cases, Lustig et al.[58] found that in 72% of patients with either chronic signs and symptoms (i.e., pertaining to a sinus tract, osseous defect, or mobility) or acute exacerbations, there was greater bone loss recorded compared to patients for whom a VRF diagnosis was made at an early stage.[58]

Despite the difficulty of diagnosing *early stage* VRFs in endodontically treated teeth, there are often several radiographic signs associated with *later stages* that are strong indicators of VRF.

The J-shaped or halo radiographic appearance of bone loss, a combination of periapical and periradicular radiolucencies, was associated with a high probability of a VRF in a double-blind radiographic study involving 102 endodontically

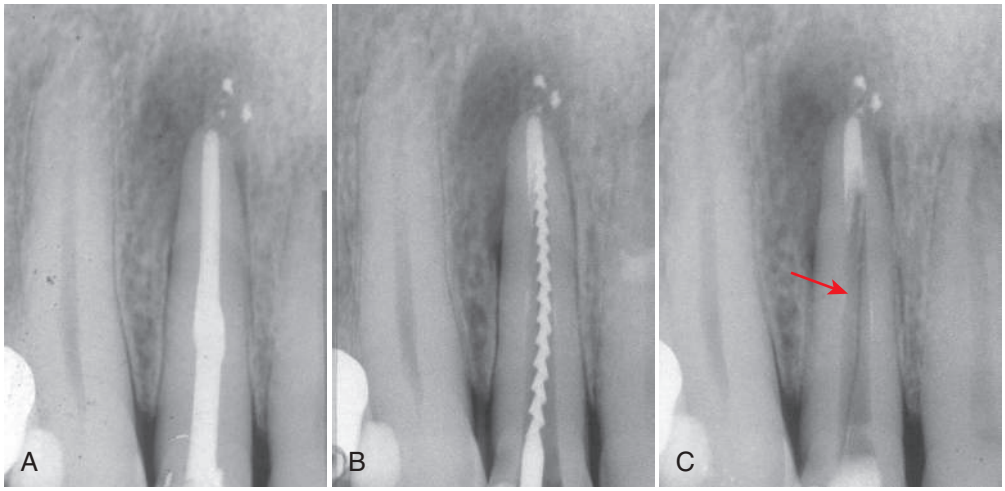


Fig. 22.18 Radiography of an empty canal: a clinical case. The right maxillary lateral incisor underwent a root canal treatment a few years ago. The patient complained about occasional pain on the palatal side. The tooth was sensitive to percussion and palpation on the palatal side. Radiography revealed a periapical radiolucency **(A).** An isolated, deep, and narrow pocket was found on the palatal side of the root. Both the patient and the referring dentist were reluctant to extract the tooth, assuming that the pocket could potentially be a sinus tract, and decided on retreatment. A radiograph taken with a file during the process **(B)** could have missed essential information that was obscured by the file, which was an evident vertical root fracture **(C).**

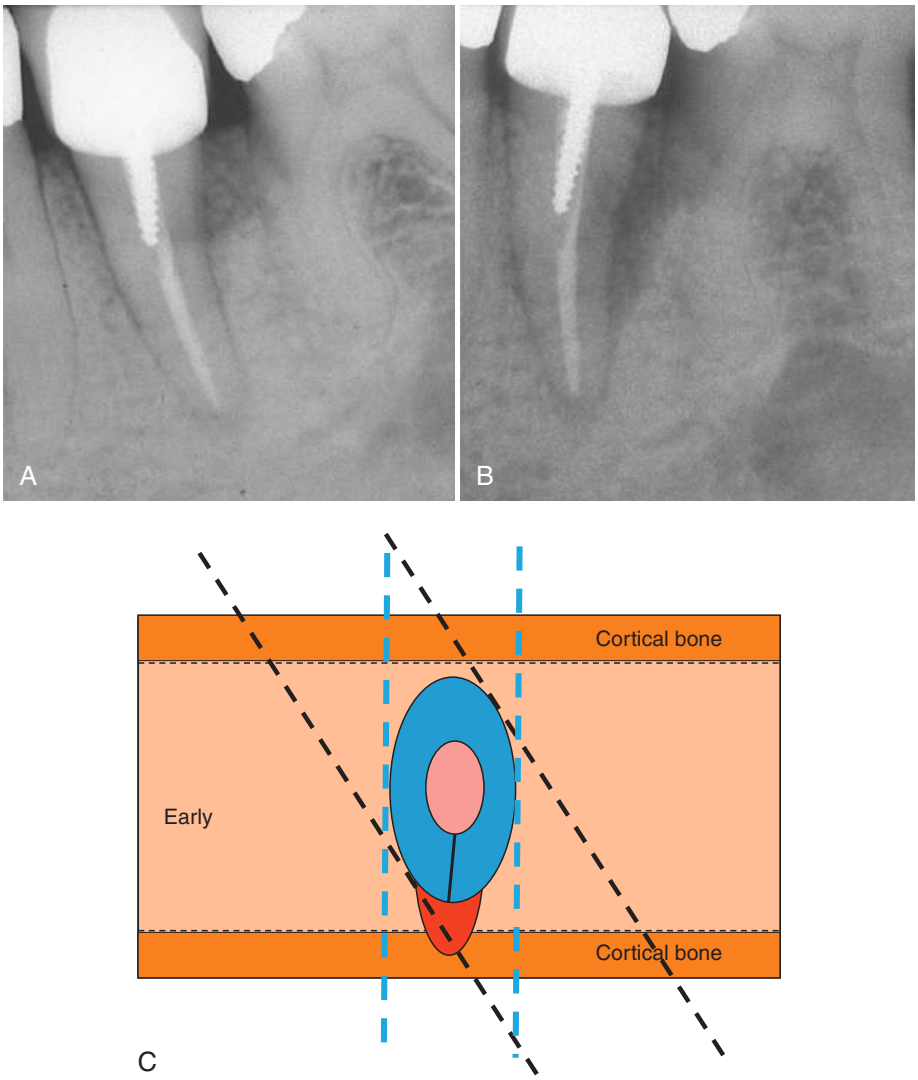


Fig. 22.19 Diagonal radiographs for detection of vertical root fracture (VRF). **A,** Orthoradial radiograph: very limited bone loss is seen. **B,** A different horizontal angulation reveals a radiolucent lesion along the root. **C,** Schematic presentation. Although a radiograph in an orthoradial direction *(blue lines)* cannot pick up the radiolucency along the root, diagonal angulation *(black lines)* may do so, as in **B.**

treated maxillary premolars (see Fig. 22.13).[83] An angular resorption of the crestal bone along the root on one or both sides, without the involvement of the periapical area, mimicking a "periodontal radiolucency" (see Fig. 22.13), was found in 14% of the cases. Tamse et al.[84] also reported the radiographic appearance of "halo" and "periodontal" radiolucencies in vertically fractured mesial roots of mandibular molars (37% and 29%, respectively). In that study, the use of these two variables, combined with bifurcation involvement (63%) and the presence of an amalgam dowel (67%), predicted fracture in 78% of the cases. Others have reported similar findings.[24,63] Despite the different sample sizes, study designs, and objectives, the most common radiographic feature in these studies was a lateral radiolucency appearing longitudinally along the root and a halo appearance.

Osseous Radiolucency Adjacent to the Root. The type of periradicular radiolucency associated with a VRF is not and should not be interpreted as a thickening of the PDL. Instead, it represents a substantial destruction of the cortical plate of the alveolar bone (see Fig. 22.12).[58] In the case of a VRF in the bucco-lingual plane, often the bone resorption is limited at early stages, and any associated radiolucency may be obscured by the superimposition of the root (see Fig. 22.12). As the bone loss increases, the radiolucency becomes greater than the dimensions of the root, allowing it to be detected more clearly in the previously mentioned manner (see Fig. 22.12). As the VRF progresses to an intermediate stage, radiographs taken at different horizontal angulations may detect bone resorption (see Fig. 22.19), whereas a single radiograph may not (see Figs. 22.12 and 22.19). This radiographic feature should be differentiated from a split tooth, in which the fracture plane is typically mesiodistal, with the bone resorption occurring in the earlier stages on the mesial or distal aspects of the root.

Radiograph of Empty Canal. As mentioned previously, the direct clinical detection of an early stage VRF from a periapical radiograph is unlikely, especially when there has been endodontic treatment. Because most VRFs are in the

bucco-lingual plane, the radiopaque obturation often obstructs the view of the hairline radiolucency of the fracture (see Fig. 22.11). When a VRF is suspected, one may initiate root canal retreatment, removing the root obturation, and exposing radiographs at two or three different horizontal angulations. The detection of a hairline radiolucency may provide a more definitive diagnosis of a VRF (see Fig. 22.18).[81]

Cone-Beam Computed Tomography in Vertical Root Fracture Diagnosis. Modern cone-beam computed tomography (CBCT) has a much smaller radiation dose compared to traditional medical spiral CT imaging, thus rendering CBCT a reasonable diagnostic tool for use in selected endodontic cases.[6,31,32,66]

One of the unique features of CBCT is its ability to study the suspected tooth and associated bone in an axial plane (i.e., the horizontal sectioning of the root). Axial views may provide detailed information regarding the cross-sectional appearance of the tooth and its surrounding bone (Fig. 22.20). Considering the resolution of the current CBCT devices, the width of an unseparated fracture may be too small to be detectable (see Fig. 22.20) (SEDENTEXCT guidelines[31]). Traditional planar, periapical radiographs are also of limited value for the *early* detection of VRFs. More specifically, bone damage or separation of the fragments is only radiographically evident at a relatively late stage. Several studies suggested that the detection of early-stage VRFs by a CBCT scan set to an axial view may be possible.[42,43,60] Yet such detection may greatly depend on the resolution of the machine (i.e., the voxel size). At a voxel size of 0.3 mm, the detection of early, unseparated VRFs is not reliable; however, when smaller voxel sizes were used in these in vitro studies, the reliability greatly increased.[42,43,60] Although the detection level of a fracture is thought to be the width of twice the voxel size of the CBCT imaging, there is presently no literature available to support this theory. Therefore, given that the smallest voxel size currently available for a CBCT device is about 0.075 mm, CBCT imaging would not be able to visualize a root fracture unless the fracture width was greater than 0.15 mm. It should also be noted that the intracanal presence of gutta-percha or a metal post often causes artifacts that make it extremely difficult to discern a VRF.[60]

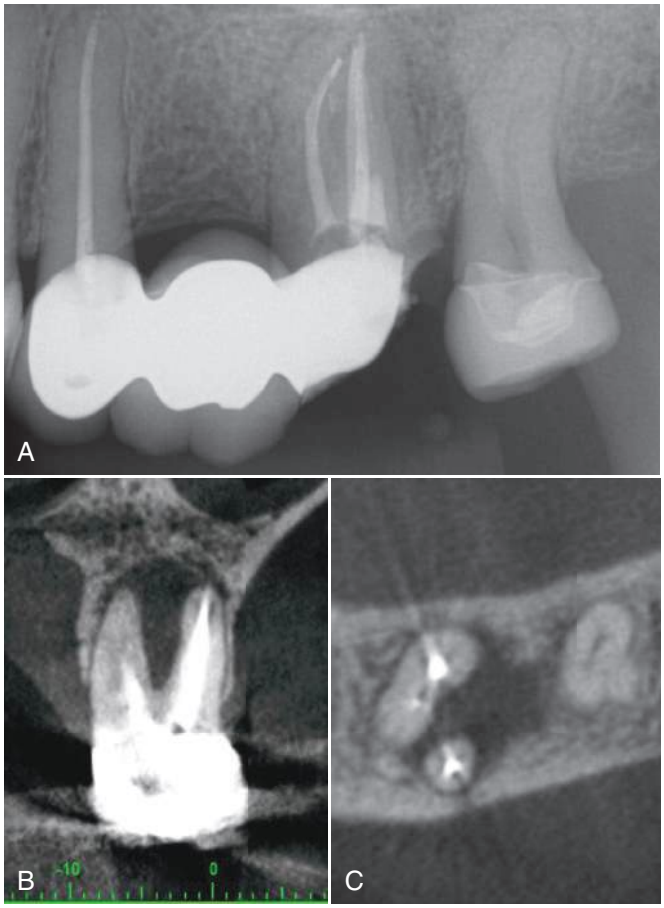


Fig. 22.20 **A,** Periapical radiograph of maxillary left first molar. **B,** Cone beam computed tomography (CBCT) coronal view revealing bone loss between the buccal and palatal roots, suggestive of a vertical root fracture. **C,** CBCT axial view revealing bone loss between the buccal and palatal roots, suggestive of a vertical root fracture. (Courtesy Dr. R. Granik.)

Although the width of early VRFs may still be below the detection level of many CBCT devices, the *early destruction of the bone* along the suspected fracture may be visible in the cancellous bone (i.e., with an axial view) at relatively early stages, whereas this early bone destruction would not be detectable in traditional two-dimensional periapical radiographs; such bone resorption may help to establish a VRF diagnosis (see Fig. 22.20).

With likely increased resolution in the near future, CBCT may become a more important diagnostic tool for the detection of VRFs. For the present, neither the most updated Joint Position Statement of the American Association of Endodontists and American Academy of Oral and Maxillofacial Radiology (2010), nor the European Society of Endodontology position statement on the use of CBCT in endodontics (2014) recommend the use of CBCT for a definitive diagnosis of VRF.[6,32]

Improvements of CBCT imaging—such as achieving a better signal-to-noise ratio, obtaining a smaller voxel size, and by applying advanced algorithms to segment fracture lines—may promise the potential to enhance the ability to detect early-stage VRFs in the future.

Exploratory Surgery. When clinical and radiographic evaluations are equivocal in detecting a suspected VRF, exploratory surgery may be indicated. In the presence of a bony lesion, especially a bony dehiscence, the raising of a full-thickness flap and removal of the granulation tissue may help to facilitate the visualization of a VRF (Fig. 22.21; also see Fig. 22.12).[81] The bone resorption pattern associated with a VRF is most often seen as a bony dehiscence, with the greater bone destruction being present on the buccal cortical plate located over the offending root. In a small percentage of the cases, a fenestration can be seen.[58] Furthermore, it has been shown that the longer a VRF-related infection persists, the greater the resulting periradicular bone destruction.[58]

ETIOLOGY

VRFs may arise from a series of factors, some of which are natural whereas others are iatrogenic, arising from dental



Fig. 22.21 Clinical image of a vertical root fracture, observed using applied dye. (Courtesy Dr. S. Floratos, Athens, Greece.)

procedures such as endodontic treatment and restorative procedures. The most common dental procedure contributing to VRFs is endodontic treatment.[11]

Most VRFs occur in endodontically treated teeth.[11,24] VRFs usually do not occur during the actual obturation of the root canal, but rather long after the procedure has been completed.[82]

The etiology of VRFs is multifactorial,[34,80] arising in the presence of one or more predisposing factors, the repeated functional or parafunctional occlusal loads which may eventually lead, over months or even years, to the development of a VRF. Predisposing factors may include natural influences, such as the anatomy of the root, or iatrogenic causes, such as the excessive forces during root canal instrumentation, excessive tooth structure removal, or excessive obturation pressure.

Natural Predisposing Factors

Shape of Root Cross Section. One of the common anatomic features shared by teeth that typically develop VRFs is an oval cross section of the root, with a bucco-lingual diameter being larger than the mesiodistal diameter.[37,41] These teeth include the maxillary and mandibular premolars, the mesial roots of the mandibular molars, and the mandibular incisors (Fig. 22.22, *A* and *B*). Such anatomy is easily observed in the axial plane of a CBCT scan (see Fig. 22.22, *A* and *B*). The fracture in these teeth typically starts in the bucco-lingual plane, specifically at the highest convexity of the oval root (see Fig. 22.22, *A* and *B*).[21,81] This conclusion, derived from large case series, is also supported by finite element analysis.[65] Such analysis clearly demonstrated strain concentration on the inner side of the remaining dentin wall at the highest convexity point (i.e., the buccal and lingual sides of the oval roots) (Fig. 22.23).[55,56]

Occlusal Factors. Excessive occlusal loads or concentration of such loads may be another natural predisposing factor for VRFs. Load concentrations, such as those caused

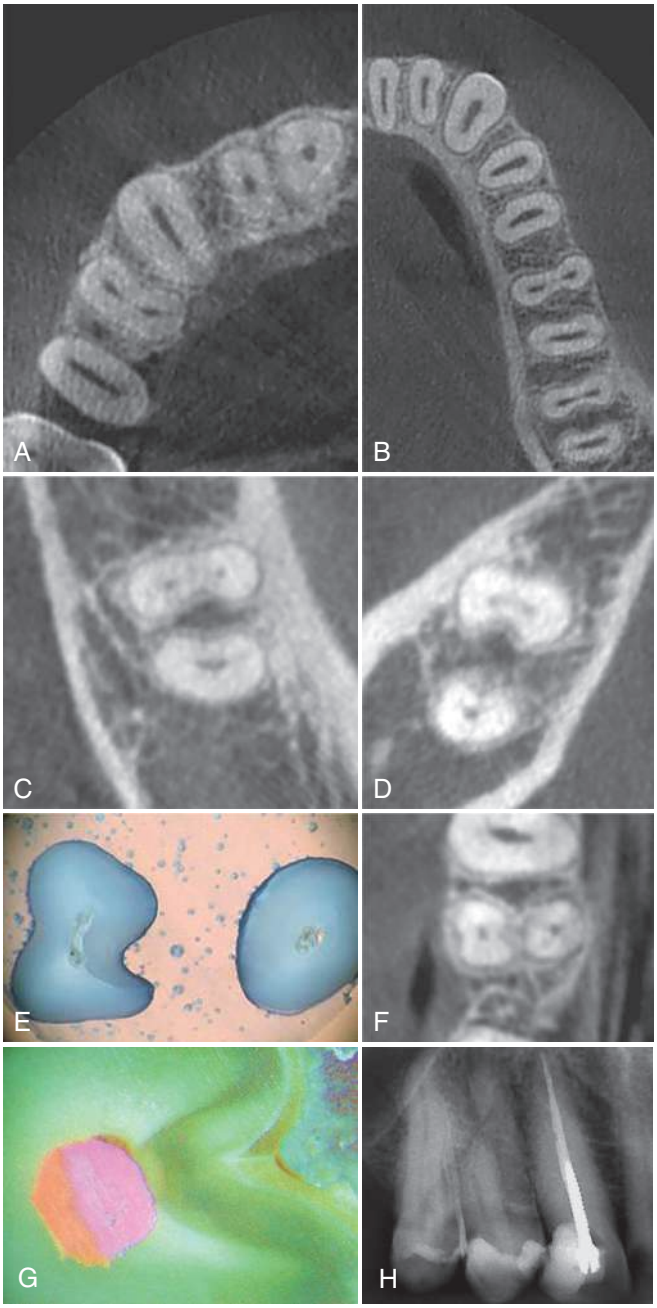


Fig. 22.22 Anatomic predisposing factors. **A,** An axial view of cone-beam computed tomography (CBCT) scan of a maxilla, revealing oval canals in the maxillary canine and second premolar. **B,** An axial view of a mandible, revealing oval canals in incisors, canine, premolars, and distal roots of the mandibular molars. Oval anatomy combined with endodontic treatment has been associated with a higher incidence of vertical root fractures (VRFs). **C** and **D,** Concavities on the distal aspect of the mesial roots of mandibular molar may establish a "danger zone" in which excessive instrumentation, combined with straightening of the canal, may result in a thinner dentin wall that may allow strain concentration. **E** and **F,** Concavities in the palatal side of the buccal root of maxillary first premolar (**E**, sections; **F**, axial view from a CBCT). These depressions may also represent a potential danger zone. Neither the concavity in **C** and **D** nor the concavity in **E** and **F** would be evident in a planar periapical radiograph. It should be noted that CBCT scans should not be used for routine screening but should be limited to the indications delineated in the joint statement of the American Association of Endodontists (AAE) and the American Academy of Oral and Maxillofacial Radiology (AAOMR).[5,6] **G,** Histology of VRF. **H,** Radiograph revealing peri-radicular bone loss. (Courtesy Dr. R. Ganik.)

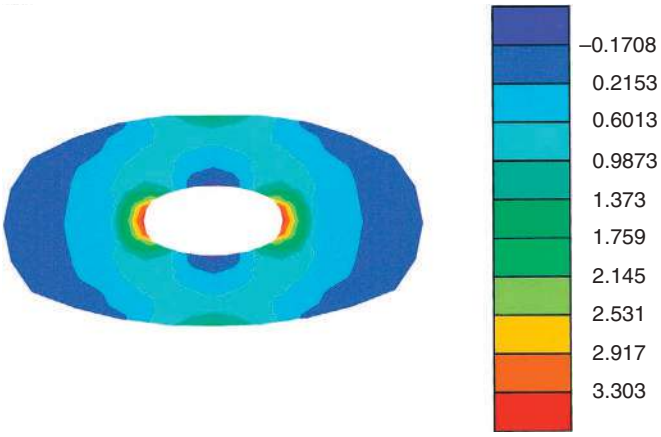


Fig. 22.23 Finite element analysis of the strain distribution in an oval root. Note the strain concentration on the inner side of the highest convexity of the remaining dentin wall. Red and orange represent areas of higher strains than blue areas. (From Lertchirakarn V, Palamara J, Messer HH: Patterns of vertical fractures: factors affecting stress distribution in the root canal, *J Endod* 29:523, 2003.)

by occlusal prematurities in maxillary premolars, and excessive occlusal forces, specifically in the case of mandibular second molars, are good illustrations.[18] In combination with other natural and iatrogenic predisposing factors, excessive occlusal loads may, over time, lead to VRFs.

Preexisting Microcracks. Preexisting microcracks may be present in the radicular dentin, likely resulting from repeated forces of mastication or occlusal parafunction.[15,64] Such fractures were also recently reported by Barreto et al.,[10] who found these microcracks present in 40% of intact maxillary incisors and canines.

Iatrogenic Predisposing Factors

Root Canal Treatment. VRFs mostly appear in endodontically treated teeth[5,81]; therefore, endodontic treatment per se may be considered an iatrogenic predisposing factor. Teeth were once thought to be more susceptible to fracturing after endodontic treatment because of a decrease in hydration.[44] However, later studies found no difference in the properties of dentin, as a material, after endodontic procedures.[47,76]

Although the physical characteristics of dentin, as a material, may not be compromised by endodontic treatment, the radicular dentin, as a structure, may be compromised by the accumulative or combined effect of several natural or iatrogenic factors associated with the endodontic treatment and the restoration of endodontically treated teeth. This may be the reason for the often-reported association of VRF with endodontically treated teeth.

It is incumbent for the clinician to recognize this effect and take efforts to minimize any steps that may contribute to the development of a root fracture during endodontic treatment.

Excessive Root Canal Preparation. Excessive root canal preparation may be a predisposing factor for VRF development.[90] In one study, cracks detected by transillumination were more frequent when the same teeth were subjected to a gradually increasing endodontic canal preparation.[90] To reduce the risk of VRFs, less invasive methods may be considered, such as minimally invasive endodontic instrumentation (see also Chapter 8).[61,67]

Microcracks Caused by Rotary Instrumentation. Shemesh et al.[77] and others[2,10,14,17,45,78,92] observed that root canal preparation using nickel-titanium rotary and reciprocating files often results in microcracks in the remaining radicular dentin (Fig. 22.24). This finding, which was originally noted for single rooted teeth, has been further supported by Yoldas et al.,[92] who studied microcrack formation from rotary files in the mesial roots of mandibular molars. Each of the rotary file systems examined in this study caused frequent microcracks in the dentin, whereas both hand instrumentation with files and the self-adjusting file (see Chapter 8) did not cause such cracks.

A finite element analysis by Kim et al.[51] supports and may explain these findings. These researchers and others reported that rotary files induce strain on the dentin, as measured in the surface layers of the root dentin, which likely exceeds the elasticity of the dentin, causing subsequent microcracks.[2,10,14,17,45,77,78,92] Additional stress, either from root obturation with lateral compaction[10,77] or by nonsurgical endodontic retreatment,[78] caused some of the microcracks to propagate and become through-and-through fractures that were indistinguishable from VRFs.[10,77,78]

The relationship of these various findings to the subsequent clinical creation of a VRF has yet to be confirmed. In any event, considering the basic principles of fracture mechanics, the creation of microcracks should at least be considered as a potential predisposing factor.

Uneven Thickness of Remaining Dentin. The instrumentation of root canals often results in uneven thickness of the remaining dentin, particularly when curved canals are straightened by instrumentation.[68] Uneven dentin thickness can also occur upon excessive instrumentation of the mesial roots of the mandibular molars or first maxillary premolars, which may exhibit a distal or mesial concavity, respectively, that is not detectable in a common planar, periapical radiograph (see Fig. 22.22, *C* and *D*). These areas, which have been referred to as "danger zones,"[30] may be characterized by a decrease in the remaining dentin thickness in which the application of internal strain may potentially lead to a fracture. The anatomic groove that is often



Fig. 22.24 Typical bucco-lingual fracture that includes the isthmus in a maxillary premolar with two canals.

found on the palatal side of the buccal root of maxillary bifurcated premolars is another example of such a hidden danger zone (see Fig. 22.22, *E*–H).[49]

Lingual access, which is commonly used in incisors, may also result in a thinner buccal wall in the apical area as compared with the lingual wall. This phenomenon may be especially pronounced when thick, and thus rigid, instruments are used excessively. When lateral compaction was applied ex vivo in a similar case, the strain concentration was recorded on the thinner buccal side of the apical part of the tooth.

The use of flexible nickel-titanium files and minimally invasive instrumentation may reduce such risks (see also Chapter 8).

Methods of Obturation. Certain obturation techniques, such as lateral compaction, involve the application of internal pressure with a spreader, which may cause strains[27,74] and subsequent propagation of microcracks into fractures across the full dentin thickness.[10,77] Other obturation methods may create less pressure, such as thermoplasticized gutta-percha, and may reduce the risk of VRFs (see Chapter 9).

Type of Spreader Used. The use of a more rigid and thick stainless-steel hand spreader may lead to increased strain in the radicular dentin and can result in an increased incidence of root fracture.[27,74] The introduction of more flexible finger spreaders, which have smaller diameter, may greatly reduce such risks.[27] Among the finger spreaders, devices composed of nickel-titanium allow insertion with less force than stainless-steel finger spreaders.[75] The nickel-titanium spreaders also allow a further reduction in the strain induced in the radicular dentin during obturation compared to traditional stainless-steel finger spreaders.[69]

Post Design. Post selection, design, and seating have a significant effect on the strain distribution in the root. Excessively long or thick posts are considered a predisposing factor for VRFs.[23,25,62] The use of posts carries an inherent risk of root fracture, particularly if excessive dentin is removed during preparation. Posts should only be used when essential for core retention and should be avoided whenever sufficient coronal tooth structure is available for the secure retention of the crown.[4,38]

Crown Design. When considering endodontically treated teeth, crowns with a ferrule margin (i.e., supported by a sound tooth structure all around and beyond the gingival margins of the core) provide better strain distribution than similar restorations that are supported by the post and core alone.[4] This design may help to avoid yet another potential predisposing factor for VRFs.[26]

Even though endodontic treatment may be necessary because of pulpal pathosis, the impact of each of the iatrogenic predisposing factors should be minimized as much as possible, as they are likely to have a cumulative deleterious effect in perpetuating VRFs.

Treatment Planning

Prevention is the key to managing VRFs. There are many predisposing factors and iatrogenic causes of these fractures, all of which should be minimized as much as clinically possible. Whereas VRFs may be present in teeth subjected to endodontic retreatment, they are rarely present in teeth that have never undergone endodontic treatment.[22] Therefore, comprehensive clinical, radiographic, and periodontal examination is imperative when evaluating any tooth that is planned for endodontic treatment or retreatment. A flexible periodontal probe is mandatory in such examinations.

When a VRF is determined to be present, extraction of the affected tooth or root is recommended as soon as possible. Any delay may increase the potential for additional periradicular bone loss and potentially compromise the placement of an endosseous implant. Attempts to "repair" a fracture by filling the crevice with a variety of restorative materials have been reported; however, none of these repairs is considered a reliable long-term solution.[8,50,54,85]

There are additional clinical and radiographic examples that can be seen of VRFs in Figs. 22.25 and 22.26.

Summary

Because of the wide variety of different types of cracks and fractures in teeth, there may be a myriad of symptoms and presentations that may appear at different stages of the crack or fracture development, making their diagnosis often difficult. The awareness that many of these symptoms represent *stages in an evolving process* may make their interpretation easier for the clinician.

The extensiveness of a crack may directly alter the prognosis assessment and treatment plan for a given tooth; the presence of a fracture may lead to the demise of a tooth and may compromise the periradicular bone. Therefore, developing a diagnosis, prognosis assessment, and treatment plan for teeth with suspected cracks and fractures is essential, with an emphasis on early detection. In addition, endodontic and restorative procedures should focus on minimizing any offending and predisposing factors that may perpetuate cracks and fractures.

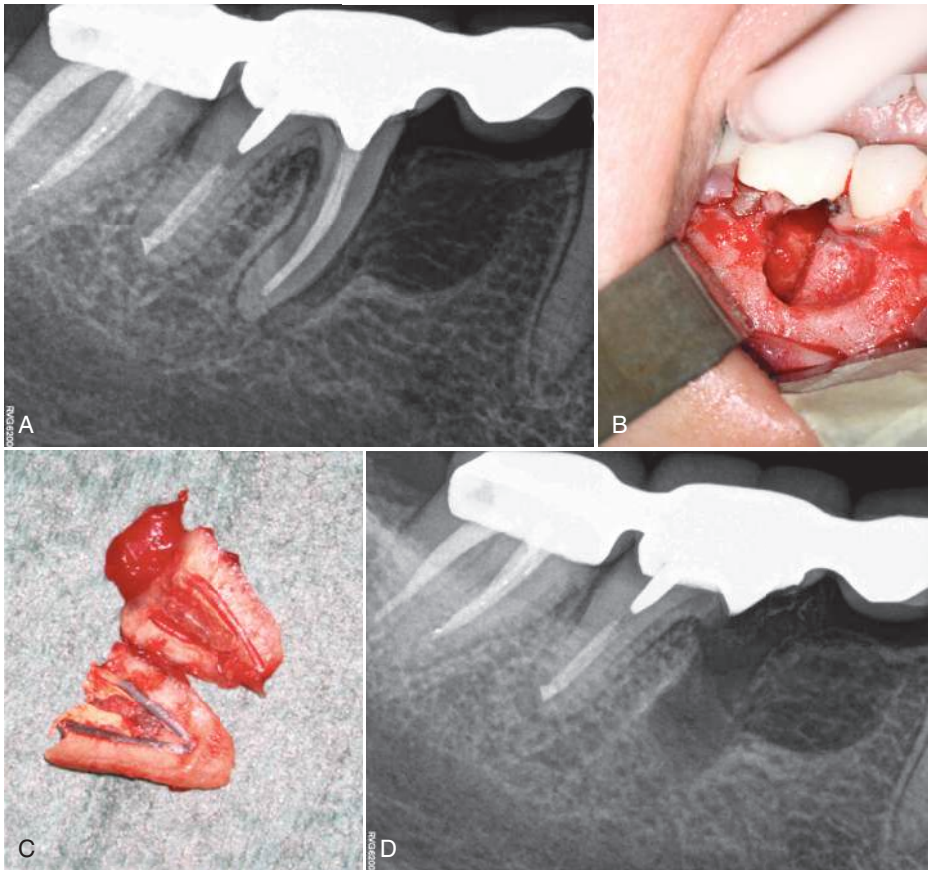


Fig. 22.25 A, Periapical radiograph revealing bone loss from crestal bone to apex. **B,** As viewed clinically after full thickness flap. **C,** Mesial root resection performed and can see how the root has split. **D,** Final radiograph after mesial root resection. (Courtesy Dr. S. Floratos, Athens, Greece.)

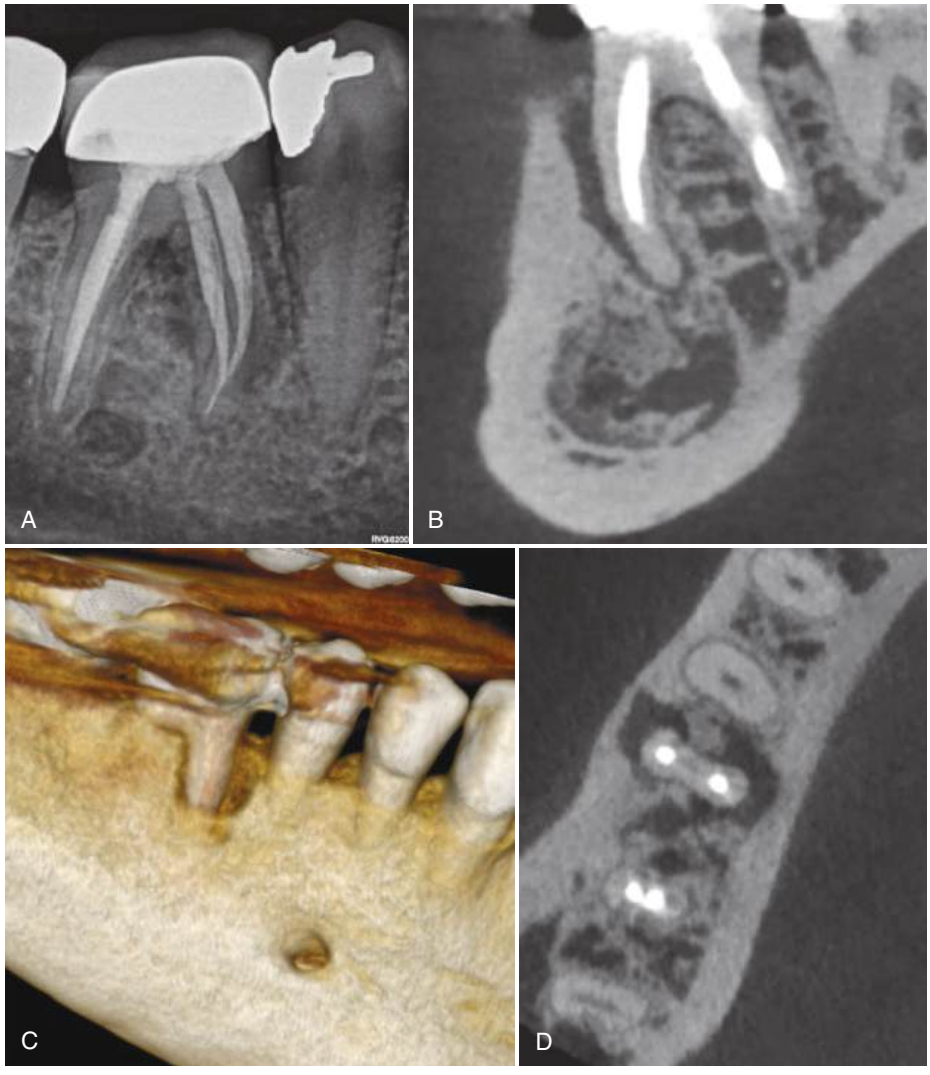


Fig. 22.26 A, Periapical radiograph of mandibular right first molar revealing suspected periradicular bone loss. **B,** Cone-beam computed tomography (CBCT) sagittal view of same tooth revealing bone loss from crestal bone to apex. **C,** CBCT anatomic view revealing mesial osseous defect. **D,** CBCT axial view revealing buccal and lingual bone loss midroot. (Courtesy Dr. R. Kasem.)

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